

NICOLÁS ANDRÉS GÓMEZ RAMÍREZ

Electronic Engineering Student | Embedded Systems & PCB Design

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EDUCATION

Pontificia Universidad Javeriana, Bogotá, Colombia
B.S. in Electronic Engineering | GPA: 4.25 / 5.0

Jan 2023 – Expected 2027

TECHNICAL SKILLS

PCB & Hardware Design: Altium Designer (schematic capture, PCB layout), LTspice, 3D design & printing

Embedded Systems: C, C++; PIC (MPLAB X, MCC), Arduino Uno / Mega 2560, Raspberry Pi (Python, GPIO); SPI, UART, I²C, PWM, MQTT; FSMs, interrupts, timing analysis

Sensors & Actuation: IR sensor, analog inputs, 12-bit SPI ADC/DAC; DC motor control (H-bridge, PWM), servo control

Digital Design & Signals: VHDL, Intel Quartus, ModelSim; MATLAB, FIR/IIR filter design

Software & Tools: Python, JavaScript/TypeScript, React, Next.js, Firebase, PostgreSQL, Git, Linux

EXPERIENCE

Teaching Assistant — Arduino 101, Pontificia Universidad Javeriana

Feb 2026 – Present

- Guide undergraduate students through sensors, actuators and peripheral devices on the Arduino platform.
- Mentor projects from system design to implementation; debug embedded C (memory management, optimization).

Teaching Assistant — Processor-Based Systems Design, Pontificia Universidad Javeriana

Jan 2025 – Jun 2025

- Designed and developed custom PCBs in Altium Designer for hands-on experimentation with course concepts.
- Resolved office-hours questions on embedded C, finite-state machines, interrupts and timing analysis.
- Supported embedded-systems and microprocessor course projects; explained digital-system architectures.

Academic Tutor — Programa Tu, Instituto Alberto Merani

Sep 2024 – Present

- Deliver one-on-one tutoring in physics, mathematics, critical thinking and language (22+ contracted hours).

HARDWARE PROJECTS

Real-Time FIR Filtering on a PIC Microcontroller | C, SPI, UART

2025

- Built a two-channel signal chain: MAX11666 12-bit ADC → order-12 and order-14 FIR filters in C → MCP4822 12-bit DAC, with ADC and DAC sharing one SPI bus.
- Wrote a UART command parser and FSM to switch filter, input and DAC routing at runtime without reflashing.

Custom Course PCBs | Altium Designer

2025

- Designed custom PCBs for hands-on work in Processor-Based Systems Design, from schematic capture to layout.

PROTEO — Raspberry Pi IoT Node | Python, MQTT, Raspberry Pi

2025

- Python service on a Raspberry Pi: publishes an IR sensor over MQTT (QoS 1) and drives a servo from MQTT commands with microsecond pulse widths (pigpio); retained Last-Will topic flags the node offline to the web dashboard.

DC Motor Control for a Conveyor Belt | Arduino, C

2025

- Controlled DC-motor direction and speed through an L298N H-bridge with PWM, a center dead zone and a minimum duty cycle that overcomes static friction; streamed telemetry over serial.

DreamSnake — Two-Player Game on Arduino Mega 2560 | C++, team of 3

2023

- Drove two MAX7219 8×8 LED matrices and a 20×4 I²C LCD; read two-player buttons; sequenced audio with an FSM.

SELECTED SOFTWARE PROJECTS

- **Voltio (2026)** — PWA for shared EV charging in a residential complex (pilot: 3 towers, 96 apartments): kWh billing from meter photos, bookings and per-tower consumption reports. JavaScript, Firebase.
- **Video_DJ (2026)** — In-progress browser loop station driven by camera-based hand and body tracking, planned in an 11-document engineering spec with a latency analysis. TypeScript.

LANGUAGES

Spanish: Native | **English:** Upper-Intermediate (B2), Berlitz English Program, 2026